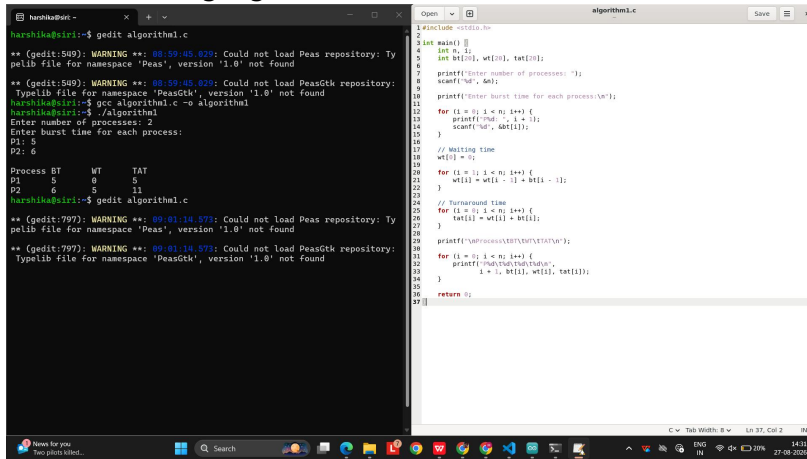


1. FCFS Scheduling algorithm

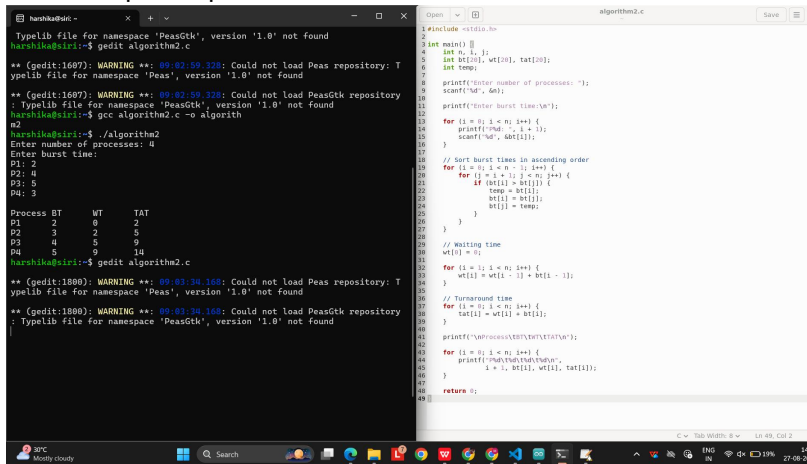


The screenshot shows a terminal window on the left and a C++ code editor on the right. The terminal displays the execution of the FCFS scheduling algorithm for 2 processes. The code editor shows the implementation of the algorithm in C++.

```
harshika@kali:~$ gcc algorithm1.c
** (gdb:509): WARNING **: 09:59:45.929: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gdb:549): WARNING **: 09:59:45.929: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@kali:~$ ./algorithm1
Enter number of processes: 2
Enter burst time for each process:
P1: 5
P2: 6
Process BT    WT    TAT
P1    5    0    5
P2    6    5    11
harshika@kali:~$ gcc algorithm1.c
** (gdb:797): WARNING **: 09:01:14.972: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gdb:797): WARNING **: 09:01:14.972: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@kali:~$ gcc algorithm2.c -o algorithm2
harshika@kali:~$ ./algorithm2
Enter number of processes: 4
Enter burst time:
P1: 2
P2: 4
P3: 5
P4: 3
Process BT    WT    TAT
P1    2    0    2
P2    3    2    5
P3    4    5    9
P4    5    9    14
harshika@kali:~$ gcc algorithm2.c -o algorithm2
** (gdb:1800): WARNING **: 09:03:24.168: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gdb:1800): WARNING **: 09:03:24.168: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

```
#include <iostream.h>
2
3 int main() {
4     int n, i, j;
5     int bt[10], wt[10], tat[10];
6     int temp;
7     printf("Enter number of processes: ");
8     scanf("%d", &n);
9     printf("Enter burst time for each process:\n");
10    for (i = 0; i < n; i++) {
11        printf("P%d: ", i + 1);
12        scanf("%d", &bt[i]);
13    }
14    // Waiting time
15    wt[0] = 0;
16    for (i = 1; i < n; i++) {
17        wt[i] = wt[i - 1] + bt[i - 1];
18    }
19    // Turnaround time
20    for (i = 0; i < n; i++) {
21        tat[i] = wt[i] + bt[i];
22    }
23    printf("Process\tBT\tWT\tTAT\n");
24    for (i = 0; i < n; i++) {
25        printf("P%d\t%d\t%d\t%d\n", i + 1, bt[i], wt[i], tat[i]);
26    }
27    return 0;
28 }
```

2. SJF_NONpreemptive

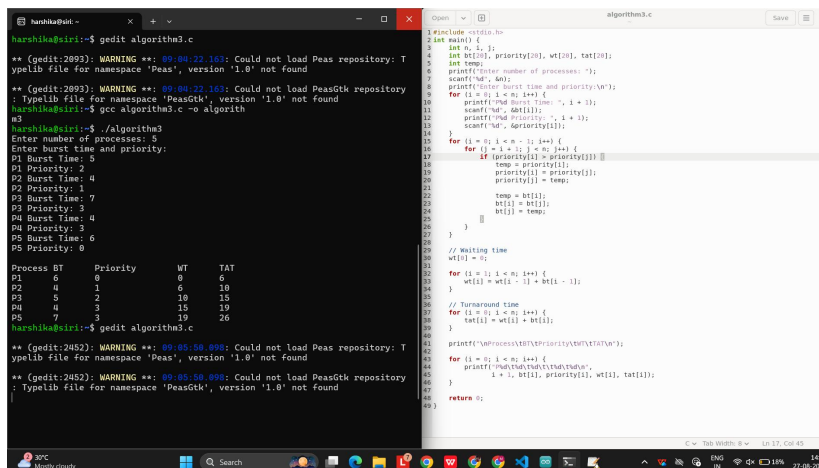


The screenshot shows a terminal window on the left and a C++ code editor on the right. The terminal displays the execution of the SJF_NONpreemptive scheduling algorithm for 4 processes. The code editor shows the implementation of the algorithm in C++.

```
harshika@kali:~$ gcc algorithm2.c
** (gdb:1697): WARNING **: 09:02:59.228: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gdb:1697): WARNING **: 09:02:59.228: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@kali:~$ gcc algorithm2.c -o algorithm2
harshika@kali:~$ ./algorithm2
Enter number of processes: 4
Enter burst time:
P1: 2
P2: 4
P3: 5
P4: 3
Process BT    WT    TAT
P1    2    0    2
P2    3    2    5
P3    4    5    9
P4    5    9    14
harshika@kali:~$ gcc algorithm2.c
** (gdb:1800): WARNING **: 09:03:24.168: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gdb:1800): WARNING **: 09:03:24.168: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

```
#include <iostream.h>
2
3 int main() {
4     int n, i, j;
5     int bt[10], wt[10], tat[10];
6     int temp;
7     printf("Enter number of processes: ");
8     scanf("%d", &n);
9     printf("Enter burst time for each process:\n");
10    for (i = 0; i < n; i++) {
11        printf("P%d: ", i + 1);
12        scanf("%d", &bt[i]);
13    }
14    // Sort burst times in ascending order
15    for (i = 0; i < n - 1; i++) {
16        for (j = i + 1; j < n; j++) {
17            if (bt[i] > bt[j]) {
18                temp = bt[i];
19                bt[i] = bt[j];
20                bt[j] = temp;
21            }
22        }
23    }
24    // Waiting time
25    wt[0] = 0;
26    for (i = 1; i < n; i++) {
27        wt[i] = wt[i - 1] + bt[i - 1];
28    }
29    // Turnaround time
30    for (i = 0; i < n; i++) {
31        tat[i] = wt[i] + bt[i];
32    }
33    printf("Process\tBT\tWT\tTAT\n");
34    for (i = 0; i < n; i++) {
35        printf("P%d\t%d\t%d\t%d\n", i + 1, bt[i], wt[i], tat[i]);
36    }
37    return 0;
38 }
```

3. Priority



The screenshot shows a terminal window on the left and a C++ code editor on the right. The terminal displays the execution of the Priority scheduling algorithm for 5 processes. The code editor shows the implementation of the algorithm in C++.

```
harshika@kali:~$ gcc algorithm3.c
** (gdb:2093): WARNING **: 09:04:22.163: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gdb:2093): WARNING **: 09:04:22.163: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@kali:~$ gcc algorithm3.c -o algorithm3
harshika@kali:~$ ./algorithm3
Enter number of processes: 5
Enter burst time and priority:
P1 Burst Time: 5
P1 Priority: 2
P2 Burst Time: 4
P2 Priority: 1
P3 Burst Time: 7
P3 Priority: 3
P4 Burst Time: 4
P4 Priority: 3
P5 Burst Time: 6
P5 Priority: 0
Process BT    Priority    WT    TAT
P1    5    0    0    5
P2    4    1    6    10
P3    7    2    10    17
P4    4    3    15    19
P5    6    3    19    25
harshika@kali:~$ gcc algorithm3.c
** (gdb:2452): WARNING **: 09:05:50.090: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gdb:2452): WARNING **: 09:05:50.090: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

```
#include <iostream.h>
2
3 int main() {
4     int n, i, j;
5     int bt[10], priority[10], wt[10], tat[10];
6     int temp;
7     printf("Enter number of processes: ");
8     scanf("%d", &n);
9     printf("Enter burst time and priority:\n");
10    for (i = 0; i < n; i++) {
11        printf("P%d: ", i + 1);
12        scanf("%d %d", &bt[i], &priority[i]);
13    }
14    // Sort processes by priority
15    for (i = 0; i < n - 1; i++) {
16        for (j = i + 1; j < n; j++) {
17            if (priority[i] > priority[j]) {
18                temp = priority[i];
19                priority[i] = priority[j];
20                priority[j] = temp;
21            }
22        }
23    }
24    // Waiting time
25    wt[0] = 0;
26    for (i = 1; i < n; i++) {
27        wt[i] = wt[i - 1] + bt[i - 1];
28    }
29    // Turnaround time
30    for (i = 0; i < n; i++) {
31        tat[i] = wt[i] + bt[i];
32    }
33    printf("Process\tBT\tPriority\tWT\tTAT\n");
34    for (i = 0; i < n; i++) {
35        printf("P%d\t%d\t%d\t%d\t%d\n", i + 1, bt[i], priority[i], wt[i], tat[i]);
36    }
37    return 0;
38 }
```

4. ROUND ROBIN

```
harshika@irc:~$ gedit algorithm4.c
** (gedit:2714): WARNING **: 09:06:55.427: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2714): WARNING **: 09:06:55.428: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@irc:~$ gcc algorithm4.c -o algorithm
harshika@irc:~$ ./algorithm
Enter number of processes: 4
Enter burst time:
P1: 6
P2: 5
P3: 1
P4: 0
Enter time quantum: 2

Process BT  WT  TAT
P1  6  5  11
P2  5  7  12
P3  1  4  5
P4  0  0  0
harshika@irc:~$ gedit algorithm4.c
** (gedit:3122): WARNING **: 09:08:17.898: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:3122): WARNING **: 09:08:17.898: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

1 #include <stdio.h>
2 int main() {
3     int n, i;
4     int bt[10], rem_bt[10];
5     int wt[10], tat[10];
6     int quantum;
7     int time = 0;
8     int done;
9     printf("Enter number of processes: ");
10    scanf("%d", &n);
11    printf("Enter burst time:\n");
12    for (i = 0; i < n; i++) {
13        printf("P%d : ", i + 1);
14        scanf("%d", &bt[i]);
15    }
16    rem_bt[i] = bt[i];
17    wt[i] = 0;
18    }
19    printf("Enter time quantum: ");
20    scanf("%d", &quantum);
21    while (1) {
22        done = 1;
23        for (i = 0; i < n; i++) {
24            if (rem_bt[i] > 0) {
25                done = 0;
26                if (rem_bt[i] > quantum) {
27                    time = time + quantum;
28                    rem_bt[i] = rem_bt[i] - quantum;
29                }
30                else {
31                    time = time + rem_bt[i];
32                    wt[i] = time - bt[i];
33                    rem_bt[i] = 0;
34                }
35            }
36            if (done == 1)
37                break;
38        }
39        for (i = 0; i < n; i++) {
40            tat[i] = wt[i] + bt[i];
41        }
42        printf("\nProcess\tBT\tWT\tTAT\n");
43        for (i = 0; i < n; i++) {
44            printf("P%d\t\t%d\t\t%d\t\t",
45                i + 1, bt[i], wt[i], tat[i]);
46        }
47        return 0;
48    }
```